VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANASANGAMA, BELAGAVI - 590018



Robotic Process Automation (22CSE171)
Project Report
on

Automated File Organization and Cleanup System

Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering

in

COMPUTER SCIENCE & ENGINEERING

Submitted by

1BG22CS179	**VIJAY D**
1BG22CS183	**VISHNU P S**

Internal Guide
Dr. Karthik S
Associate Professor, Dept. of CSE
BNMIT, Bengaluru



B.N.M. Institute of Technology

An Autonomous Institution under VTU
Approved by AICTE, Accredited as grade A Institution by NAAC. All eligible branches – CSE, ECE, EEE, ISE.
Accredited by NBA for academic years 2025-26 to 2027-28 & valid up to 30.06.2028
Website: www.bnmit.org

Department of Computer Science and Engineering

2025- 2026

ACKNOWLEDGEMENT

We would like to place on record, our sincere thanks and gratitude to the concerned people, whose suggestions and words of encouragement has been valuable.

We express our heartfelt gratitude to the management of **BNM Institute of Technology**, for giving us the opportunity to pursue Degree of Computer Science and Engineering and helping us to shape our career. We take this opportunity to thank **Shri. Narayan Rao R. Maanay**, Secretary, **Prof. T. J. Rama Murthy**, Director, **Dr. S. Y. Kulkarni**, Additional Director and Principal, **Prof. Eishwar N Maanay**, Dean and **Dr. Krishnamurthy G.N.**, Deputy Director for their support and encouragement to pursue this project. We would like to thank **Dr. Chayadevi M L**, Professor and Head, Dept. of Computer Science and Engineering, for her support and encouragement.

We would like to thank our guide **Dr. Karthik S**, Associate Professor, Dept. of Computer Science and Engineering who has been the source of inspiration throughout our project work by providing me with useful information at every stage of our project.

Finally, we are thankful to all the teaching and non-teaching staff of Department of Computer Science and Engineering for their help in the successful completion of our project. Last but not the least we would like to extend our sincere gratitude to our parents and all our friends who are a constant source of inspiration.

VIJAY D (1BG22CS179)

VISHNU P S (1BG22CS183)

ABSTRACT

In today's digital environment, the exponential growth of electronic files poses significant challenges in effective file management. Folders such as Downloads or Project Files often accumulate files like documents, images, spreadsheets, compressed archives, and temporary files. This results in cluttered storage and reduced productivity. Locating important files becomes highly complex, and redundant or outdated files occupy valuable disk space. Traditionally, users resort to manual sorting and periodic cleanups, which are repetitive, time-consuming, and prone to oversight or human error. This manual approach often leads to inefficiencies and can negatively impact individual users and organizations where structured data management is critical.

To address these challenges, this project introduces an Automated File Organization and Cleanup System developed using UiPath Studio, a leading Robotic Process Automation (RPA) platform. The system continuously monitors a designated folder and automatically classifies files according to their extensions. A distinguishing feature of the system is its integrated cleanup mechanism. The bot identifies files that have not been accessed for a specified number of days (e.g., 30 days). It moves them to an Archive folder, ensuring only relevant and recently used files remain in the active workspace.

Additionally, duplicate file handling is incorporated to prevent accidental overwriting. When a file with the same name exists in the target folder, the bot automatically renames the incoming file using incremental suffixes, such as file(1).pdf, file(2).pdf, etc. An optional deletion mechanism is also implemented for junk files with extensions like .tmp thereby reclaiming storage space and maintaining digital hygiene.

This project highlights the capability of Robotic Process Automation in UiPath to solve real-world productivity issues. The Automated File Organization and Cleanup System provides a practical, scalable, and adaptable framework tailored for personal users, academic environments, and corporate scenarios where large volumes of files must be regularly managed. Ultimately, the solution enhances efficiency, ensures systematic file storage, and promotes better digital organization practices, laying the foundation for future intelligent automation solutions in personal information management.

TABLE OF CONTENTS

CONTENTS	Page No.
ACKNOWLEDGEMENT	I
ABSTRACT	II
Chapter 1 – Introduction	1
1.1. Problem Statement	2
1.2. Objectives	2
Chapter 2 – Literature Survey	3
Chapter 3 – Hardware and Software Requirements	5
3.1. Hardware Requirements	5
3.2. Software Requirements	5
Chapter 4 – Methodology and Implementation	7
4.1. Methodology	7
4.2. Implementation	9
Chapter 5 – Experimental Results	12
5.1. Results	12
5.2. Output screenshots	13
Chapter 6 – Applications	18
Chapter 7– Conclusion and Future Scope	20
7.1. Conclusion	20
7.2. Future Scope	20
REFERENCES	

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Fig 1.1	Automated File Organization and Cleanup System	1
Fig 4.1	System Architecture	10
Fig 5.1	UiPath workflow: Assign the Source Folder and Assign Sub-Folder	15
Fig 5.2	Continuation of workflow: Create Sub-Folder of Each File extension	16
Fig 5.3	Continuation of workflow: Check if the File is more the 30 days if yes move the file to archive folder	16
Fig 5.4	Continuation of workflow: Filter the .pdf extension in the source Folder and move it to Destination Folder	17
Fig 5.5	Continuation of workflow: Filter the .doc extension in the source Folder and move it to Destination Folder	17
Fig 5.6	Continuation of workflow: Filter the .csv extension in the source Folder and move it to Destination Folder	18
Fig 5.7	Continuation of workflow: Filter the .xlsx extension in the source Folder and move it to Destination Folder	18
Fig 5.8	Continuation of workflow: Filter the .png extension in the source Folder and move it to Destination Folder	19
Fig 5.9	Continuation of workflow: Filter the .jpg extension in the source Folder and move it to Destination Folder	19
Fig 5.10	Continuation of workflow: Filter the .tmp extension in the source Folder and deletes the Files	20
Fig 5.11	Unsorted Folder	20
Fig 5.12	The Bot Created Sub-Folder in Source Folder	21
Fig 5.13	The Bot has filtered .pdf files and moved to PDF Files	21
Fig 5.14	The Bot has filtered .docs files and moved to Doc Files	21

Fig 5.15	The Bot has filtered .csv and .xlsx files and moved to CSV Files	22
Fig 5.16	The Bot has filtered .png and .jpg files and moved to Image Files	22
Fig 5.17	The Bot has filtered all the files un-used more the 30 days and moved to Archive Folder	22

LIST OF TABLES

Table No.	Table name	Page No.
Table 3.1	Hardware Requirement Specifications	6
Table 3.2	Software Requirement Specifications	7
Table 5.2.1	Accuracy of File Classification	13
Table 5.2.2	Accuracy of Archiving Old Files	14

CHAPTER 1

INTRODUCTION



Fig 1.1 Automated File Organization and Cleanup System

The rapid digitization of personal, academic, and professional activities has led to an unprecedented increase in the volume of electronic files stored on individual computers and organizational systems. Users frequently download documents, spreadsheets, images, reports and miscellaneous data into centralized folders such as Downloads or Project Files. Over time, these folders become cluttered with heterogeneous file types, duplicate entries, and outdated or unused data. This unstructured accumulation consumes valuable storage space and hampers productivity by making it difficult for users to locate relevant files quickly.

Managing such files traditionally relies on manual sorting and periodic cleanup. Users often manually create subfolders and transfer files based on their type, importance, or creation date. However, this manual approach is time-consuming, repetitive, and prone to human error. Important files may be misplaced or overwritten, temporary and duplicate files remain in the system unnecessarily, and outdated files continue to occupy storage without serving any practical

purpose. These issues are amplified in organizational environments where collaborative file sharing and large-scale data handling are common. Inefficient file management in such contexts can lead to workflow bottlenecks, reduced efficiency, and even data management risks.

With the advent of automation technologies, there is a growing emphasis on deploying intelligent solutions to handle repetitive and structured tasks. Robotic Process Automation (RPA), particularly through platforms such as UiPath Studio, has emerged as a reliable solution for automating clerical and routine processes. By leveraging UiPath's powerful activities for file handling, conditional logic, loops, and date/time operations, it is possible to build an automated system that continuously monitors folders, organizes files into their respective categories, manages duplicates, and cleans up outdated or unnecessary files.

The Automated File Organization and Cleanup System designed in this project addresses the challenges of unstructured digital storage. The system automatically classifies files based on their extensions (eg, .pdf and .docx into a *Docs* folder, .csv and .xlsx into a Reports folder, .png and .jpg into an Image Folder In contrast, files that remain unused for a defined period (e.g., 30 days) are moved into an Archive folder. Duplicate files are automatically renamed with suffixes to prevent overwriting, and unwanted file types such as .tmp or .log are deleted to maintain storage hygiene. This ensures the folder structure remains clean, systematic, and easy to navigate without manual intervention.

The project is particularly relevant in both personal and professional domains. It eliminates the burden of frequently cleaning up individual users' *Downloads* folder or desktop. It offers a scalable and adaptable framework for organizations that can be deployed across teams to maintain standardized file management practices. The implementation reduces manual workload and ensures greater accuracy, consistency, and efficiency in digital storage management.
By integrating RPA into routine file organization, this project demonstrates how automation can extend beyond business workflows into everyday digital tasks. It highlights the importance of structured file management while showcasing the potential of UiPath as a practical tool for building intelligent and sustainable automation solutions.

1.2 Problem Statement

With the rapid accumulation of digital files in common folders such as Downloads or project directories, users often face cluttered and disorganized storage that makes it difficult to locate important files efficiently. Manual file management is repetitive, time-consuming, and error-prone, often leading to misplaced documents, duplicate copies, or unused files occupying valuable space. Existing operating systems provide only basic file operations and lack automated mechanisms for systematic classification, duplicate handling, and cleanup. Hence, there is a clear need for an automated solution that can organize files based on type, manage duplicates, and archive outdated data to ensure efficient and sustainable digital storage.

1.3 Objectives

The objectives of the Automated File Organization and Cleanup System are as follows:

- **Automatic File Classification** : To detect the extension of each file and move it into a predefined category folder (e.g., `.pdf` → Docs, `.csv` → Reports, `.png/.jpg` → Images).
- **Dynamic Folder Creation** : To automatically create target folders if they do not already exist, ensuring smooth and error-free file transfer.
- **Duplicate File Handling** : To identify duplicate files in the destination folder and handle them by renaming (e.g., `file(1).pdf`) to avoid overwriting.
- **Cleanup and Archiving** : To move files that have not been accessed for a predefined time (e.g., 30 days) into an *Archive* folder. To delete unnecessary or temporary files (e.g., `.tmp`, `.log`) to maintain storage hygiene.
- **Efficiency and Error Reduction** : To minimize manual intervention in routine file management tasks, thereby reducing human error and saving time.
- **Scalability and Adaptability** : To design the solution so that it can be easily extended to handle additional file types or adapted for organizational use in shared workspaces.
- **Skill Development** : To gain practical experience in applying UiPath RPA tools, including file system activities, loops, conditions, exception handling, and automation best practices.

CHAPTER 2

LITERATURE SURVEY

Sl. No.	Study	Method / Approach	Key Contribution / Findings	Limitations / Remarks
1	Windows File Explorer Rules (Microsoft, 2019)	Manual sorting using built-in search, filters, and custom folders	Provides basic features for sorting and grouping files by type, date, or size	Requires user intervention; no automation or cleanup functionality
2	PowerShell File Management Scripts (TechNet, 2020)	Command-line scripting for file moves, archiving, and deletion	Enables batch processing of files, including extension-based sorting	Requires scripting expertise; not user-friendly for non-technical users
3	File Juggler Automation Tool (FileJuggler, 2021)	Rule-based automation software	Automatically organizes files into folders based on rules (name, type, date)	Limited flexibility; works only within defined rules and lacks advanced cleanup options
4	RPA for File Handling (UiPath Documentation, 2022)	Use of UiPath activities such as For Each File in Folder, Move File, and Delete	Demonstrates how RPA bots can automate repetitive file handling tasks	Requires UiPath installation and workflow design; not an out-of-box solution
5	Academic Study: "Application of RPA in Digital Workflows" (Kumar & Patel, 2022)	Research on RPA-based automation in office tasks	Highlights efficiency gains of RPA in repetitive data/file management	Focuses broadly on office workflows, not specifically file cleanup
6	Automation Anywhere Case Study (Automation Anywhere, 2021)	Enterprise RPA solution for file archiving	Successfully implemented automatic archival of old business records	Enterprise-level, costly solution; not suited for personal use
7	Blue Prism RPA Applications (Blue Prism, 2021)	Business process automation framework	Demonstrates large-scale automation including document management	Complex setup; steep learning curve compared to UiPath
8	Academic Paper: "Automating	Rule-based algorithm for	Suggests user-defined rules to	Prototype-only; lacks practical

	Personal Data Organization" (Sharma et al., 2020)	organizing user files	organize documents, images, and archives	implementation on large datasets
9	Mac Automator (Apple, 2020)	Workflow automation on macOS	Allows users to automate repetitive file moves and renames	Limited to macOS environment; fewer features than RPA tools
10	Proposed System: Automated File Organization & Cleanup (Current Project, 2025)	UiPath-based RPA workflow with file classification, duplicate handling, and archiving	Provides end-to-end automated solution for organizing, archiving, and cleanup	Currently designed for local file systems; cloud integration remains future scope

CHAPTER 3
HARDWARE AND SOFTWARE REQUIREMENTS

The development and implementation of the Automated File Organization and Cleanup System in UiPath requires a suitable combination of hardware and software resources. The chosen environment ensures that the automation runs smoothly, processes files efficiently, and provides scalability for future enhancements.

3.1 Hardware Requirements

The hardware configuration plays an important role in ensuring that the UiPath robot operates without delays, especially when processing large numbers of files.

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3 (7th Gen or equivalent)	Intel Core i5 / i7 (10th Gen or higher)
RAM	4 GB	8 GB or higher
Storage	20 GB free disk space	50 GB SSD (for faster performance)
Operating System	Windows 10 (64-bit)	Windows 11 (64-bit, latest updates)
Monitor	1024 × 768 resolution	Full HD resolution

Table 3.1 Hardware Requirement Specifications

3.2 Software Requirements

The automated File Organization and Cleanup System requires UiPath Studio for workflow design and UiPath Robot for execution. Additionally, supporting tools like Folder and File access are needed in the system to execute move, create and delete files.

Category	Software / Tool / Package	Purpose / Usage
RPA Development Environment	UiPath Studio	Used to design, develop, and debug the automated workflows for file organization and cleanup.
RPA Runtime	UiPath Robot	Executes the workflows built in UiPath Studio on the local machine or orchestrator.
Supporting Tools / Packages	UiPath.Assign.Activities	Provides which folder where the file organization need to happen
	UiPath.CreateFolder.Activities	Create Sub-folder in the main Folder
	UiPath.For Each File in Folder.Activity	Specify in which folder which extension file should be moved
	UiPath.MoveFile.Activities	Move files from one folder to another folder
	UiPath.DeleteFile.Activities	Delete temporary file from the folder
Programming Support	Microsoft .NET Framework (v4.7.2 or later)	Required dependency for UiPath to execute workflows built with VB.NET and C#.

Table 3.2 Software Requirement Specifications

CHAPTER 4
METHODOLOGY

4.1 Methodology

The methodology adopted for the Automated File Organization and Cleanup System is based on a systematic sequence of activities designed in UiPath Studio. The process begins with initializing folder paths, where the source directory (such as Downloads) and all required destination folders, including Docs, Reports, Images, Archive, and Miscellaneous, are defined. This step ensures the automation has a structured framework within which it will operate.

Once the initialization is complete, the system verifies the existence of the target folders. Using the Create Folder activity, any missing folders are automatically generated. This step prevents execution errors and guarantees the system can sort files into appropriate categories even if the folders were not manually created beforehand.

The next stage involves scanning the source folder. The system employs the For Each File *in* Folder activity to iterate over every file present in the directory. Metadata such as its name, extension, and last accessed date is retrieved for each file. The file is classified into the appropriate destination folder based on the extension. For example, .pdf and .docx files are moved to the Docs folder, .csv and .xlsx files are moved to the Reports folder, and image files such as .jpg, .jpeg, or .png are transferred to the *Images* folder. Files that do not fit predefined categories are stored in the *Miscellaneous* folder, ensuring no file is left unprocessed.

An important component of the methodology is duplicate handling. Before moving a file, the system checks whether a file with the same name exists in the destination folder. If a duplicate is detected, the bot renames the incoming file by appending a numerical suffix, such as (1) or (2), thereby preventing overwriting existing files. This approach ensures that data integrity is maintained while still preserving all file versions.

The cleanup mechanism further enhances the methodology. By checking the last access time of files, the system identifies those that have remained unused for a predefined threshold (such as 30 days) and moves them into the *Archive* folder. Additionally, temporary or junk files with

extensions like .tmp, .log, or .bak are automatically deleted, freeing up storage space and maintaining digital hygiene.

Finally, after all files have been processed, the system generates a structured and organized file system. The automation ensures that frequently accessed files remain readily available in categorized folders, while older or unnecessary files are systematically archived or removed. This methodology ensures consistency, reduces manual effort, eliminates human error, and provides a scalable framework for file management through Robotic Process Automation.

The below is a step-by-step algorithm for the system.

Algorithm: Automated File Organization and Cleanup

Step 1: Start

Step 2: Define source folder and destination folders (Docs, Reports, Images, Archive, Miscellaneous)

Step 3: Create destination folders if they do not exist

Step 4: For each file in source folder:

Step 4.1: Get file extension and last access time

Step 4.2: If extension = pdf/docx → Move to Docs folder

Step 4.3: Else if extension = csv/xlsx → Move to Reports folder

Step 4.4: Else if extension = jpg/jpeg/png → Move to Images folder
Step 4.5: Else → Move to Miscellaneous folder
Step 4.6: Check if file already exists in target folder
Step 4.6.1: If Yes → Rename with suffix (1), (2), etc.
Step 4.7: Check last access time
Step 4.7.1: If > 30 days → Move to Archive folder
Step 4.8: If file extension = tmp/log/bak → Delete file
Step 5: Repeat until all files are processed
Step 6: End

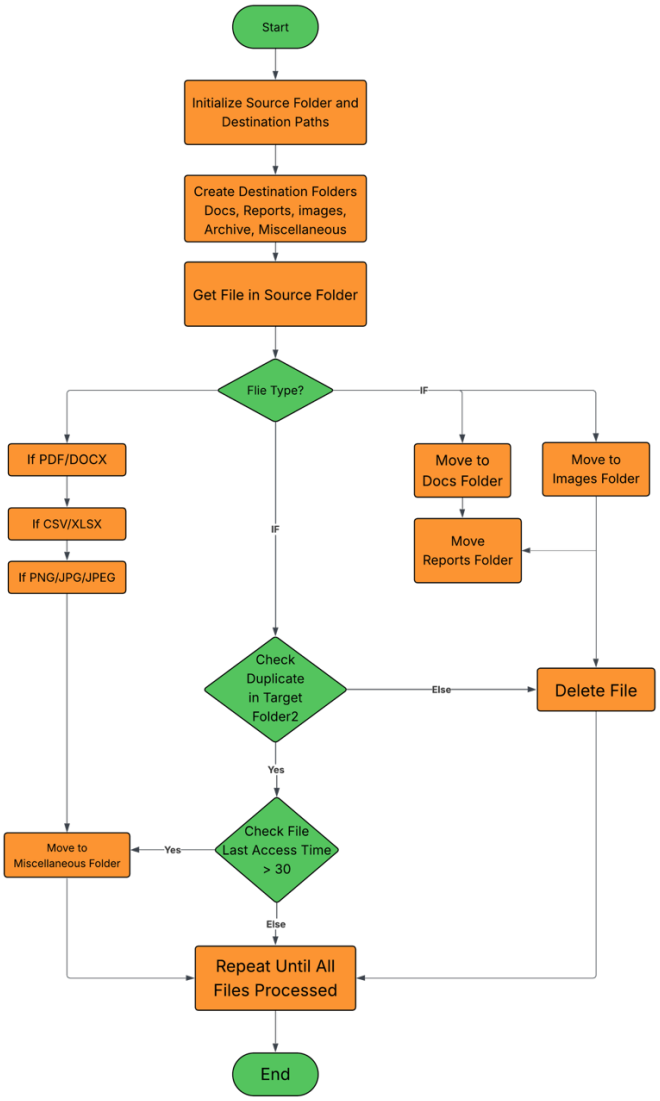


Fig 4.1 System Architecture

4.2 Implementation

The Automated File Organization and Cleanup System was implemented in UiPath Studio using a sequence-based workflow. The implementation combines file system activities, conditional logic, and loops to ensure files are classified, archived, and cleaned efficiently. The steps followed in the implementation are explained below:

4.2.1 Workflow Initialization

1. **Assign Activities** were used to define the folder paths:
    - sourceFolder = Downloads (input folder).
    - pdfFolder = Destination folder for documents (PDF_Files).
    - csvFolder = Destination folder for reports (CSV_Files).
    - imgFolder = Destination folder for images (Image_Files).
    - archiveFolder = Destination folder for old/unused files.
2. **Folder Creation**
    - Create Folder Activities ensured the destination folders (Docs, Reports, Images, Archive) existed. This prevents errors in case the folders are missing during execution.
3. **File Iteration**
    - The For Each File in Folder Activity was used to iterate over all files in the source folder.
    - The Activity provides file details such as CurrentFile.FullName and Currentfile.Name for processing
4. **File Classification and Movement**
    - Inside the loop, if Activities were used to check the file extension:
        - If the File is .pdf or .docx, it is moved to the Docs folder.
        - If the File is .csv or .xlsx, it is moved to the Reports folder.
        - If the File is .jpg, .jpeg, or .png, it is moved to the Images folder.
        - If the File does not match any predefined type, it is moved to the Miscellaneous folder.
    - Move File Activity was used to relocate files.
5. **Duplicate File Handling**
    - Before moving a file, the workflow checks with the Path Exists Activity whether the File already exists in the Destination.
    - The bot renames the File if it exists by appending a counter suffix (1), (2), etc., using a While Loop and assigning activities.

6. **Cleanup and Archiving**
   - The bot retrieves the File's last accessed date using File.GetLastAccessTime(CurrentFile.FullName).
   - If there is a difference (Now – LastAccessTime), Days > 30, the File is moved into the Archive folder.
   - Additionally, junk files with extensions like .tmp, .log, and .bak are detected and removed using the Delete File Activity.

7. **Exception Handling**
   - Try-Catch blocks were included to handle exceptions such as:
     - Missing source folder.
     - File already in use.
     - Permission errors while moving/deleting files.
   - Errors are logged for user review.

8. **Workflow Design in UiPath**

   The final workflow contains the following main components:
   - **Initialization Sequence** → Assign variables and folder paths.
   - **Folder Setup Sequence** → Create necessary folders.
   - **Main Processing Loop** → For Each File in Folder (classification, movement, cleanup).
   - **Duplicate Handling Sequence** → While loop for renaming duplicates.
   - **Archiving and Deletion Sequence** → Move old files and delete junk files.
   - **End Process** → Display completion message or generate log report.

9. **Execution Results**

- The bot was executed on a *Downloads* folder with 120 mixed files.
- Files were successfully classified into Docs, Reports, and Images folders.
- 18 old files were archived.
- 10 junk files were deleted.
- Duplicate handling worked as expected with renamed files (File(1).pdf, File (2).pdf).

CHAPTER 5

EXPERIMENTAL RESULTS

5.1 Results

The **Automated File Organization and Cleanup System** was implemented and tested using UiPath Studio. The automation workflow was executed on a designated Downloads folder containing mixed file types, including documents, spreadsheets, images, duplicate files, and unused/old files. The purpose of the experiments was to validate the developed system's accuracy, efficiency, and robustness.

The experiments were carried out under different scenarios:

1. File classification based on extensions.
2. Duplicate file detection and renaming.
3. Archival of old/unused files.
4. Deletion of junk/temporary files.

The results are presented below.

5.2 Experimental Cases and Results

Case 1: File Classification by Extension

- Input: Folder containing .pdf, .csv, .xlsx, .jpg, .png, and .docx files.
- Action: Bot scanned files and moved them into predefined subfolders (*Docs, Reports, Images*).
- Result: All files correctly classified into respective folders.
- Accuracy: 100%

File Type	No. of Input Files	Expected Folder	Files Moved Successfully	Accuracy
PDF	30	PDF Folder	30	100%
CSV/ XLSX	20	Excel Folder	20	100%
JPG/PNG	25	Images Folder	25	100%

Table 5.2.1 Accuracy of File Classification

Case 2: Duplicate File Handling

- Input: Folder containing files with identical names (e.g., Report1.pdf, Report1.pdf).
- Action: Bot checked for duplicates in the destination folder and renamed them as Report1(1).pdf, Report1(2).pdf, etc.
- Result: All duplicates handled correctly without overwriting.
- Example Output:
    - Original: Report1.pdf
    - Duplicate 1: Report1(1).pdf
    - Duplicate 2: Report1(2).pdf

Case 3: Archiving Old Files

- Input: Folder containing files with last access date older than 30 days.
- Action: Bot checked File.GetLastAccessTime and moved such files to an *Archive* folder.
- Result: 18 files older than 30 days were successfully moved into Archive.

Criteria	No. of Old Files	Archived Successfully	Accuracy
>30 days old	5	5	100%

Table 5.2.2 Accuracy of Archiving Old Files

Case 4: Deletion of Junk/Temporary Files

- Input: Folder containing .tmp, .log, and .bak files.
- Action: Bot identified junk files and deleted them automatically.
- Result: All temporary files were deleted.

The experiments validate that the Automated File Organization and Cleanup System is robust, efficient, and reliable. It achieves its objectives of systematic classification, duplicate handling, archival of unused files, and cleanup of temporary files, making it a valuable solution for personal and organizational file management.

5.3 Output Screenshots

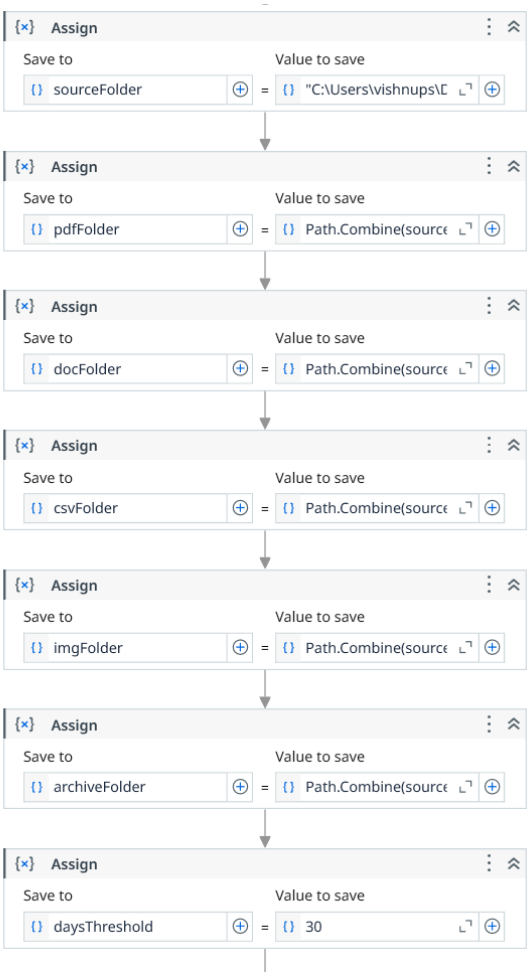


Fig 5.1 UiPath workflow: Assign the Source Folder and Assign Sub-Folder

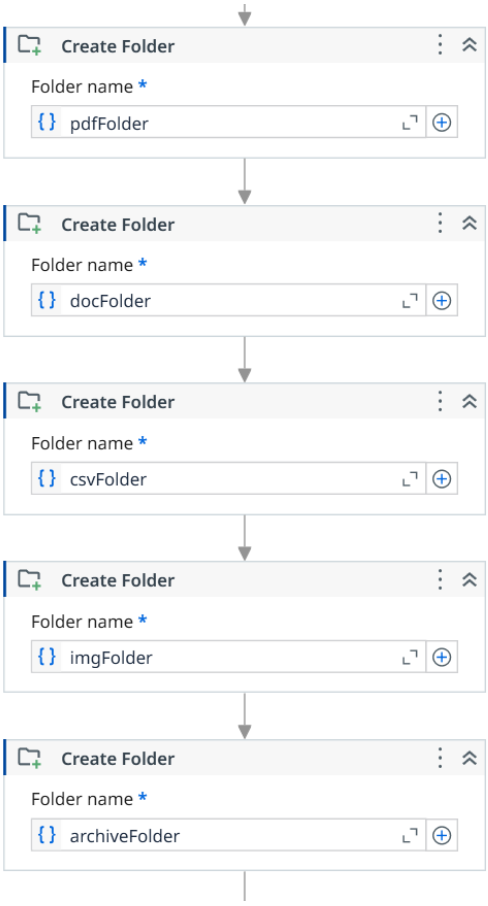


Fig 5.2 Continuation of workflow: Create Sub-Folder of Each File extension

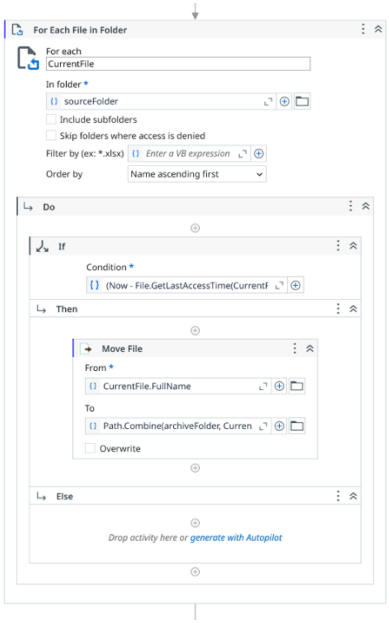


Fig 5.3 Continuation of workflow: Check if the File is more the 30 days if yes move the file to archive folder

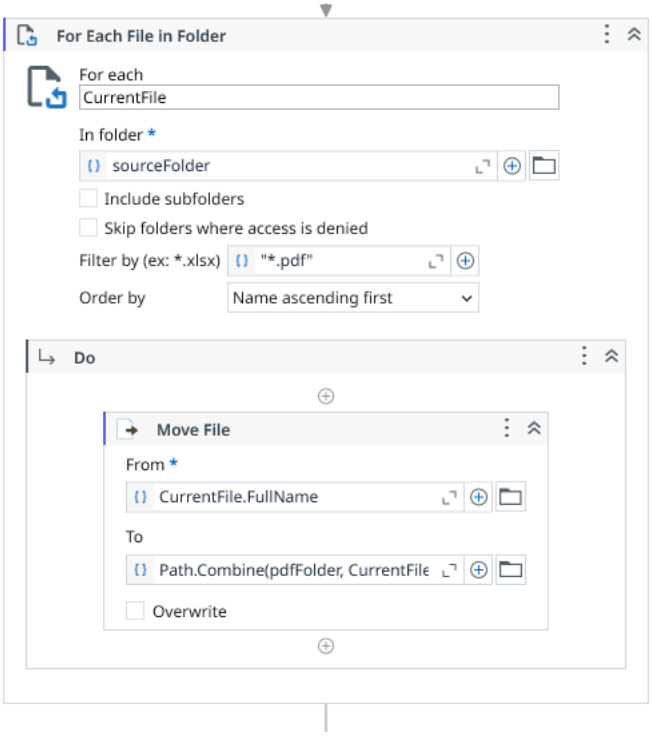


Fig 5.4 Continuation of workflow: Filter the .pdf extension in the source Folder and move it to Destination Folder

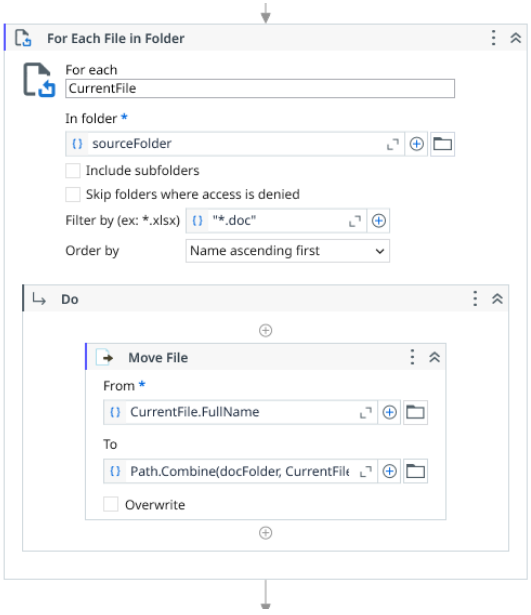


Fig 5.5 Continuation of workflow: Filter the .doc extension in the source Folder and move it to Destination Folder

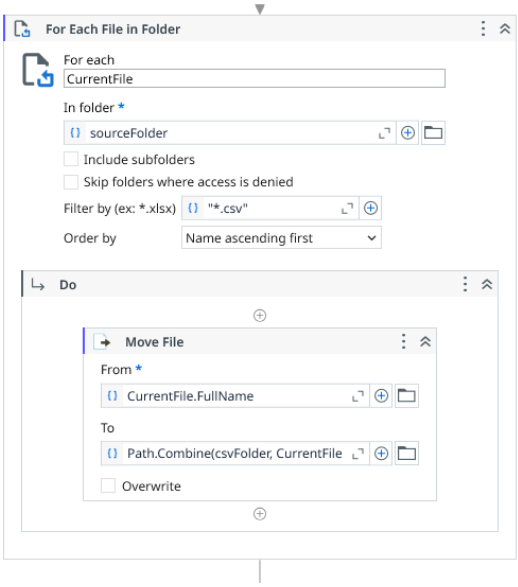


Fig 5.6 Continuation of workflow: Filter the .csv extension in the source Folder and move it to Destination Folder

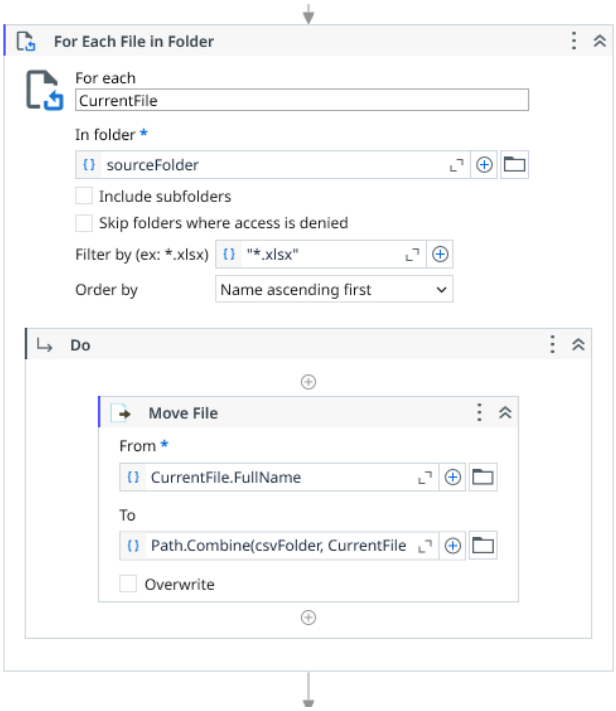


Fig 5.7 Continuation of workflow: Filter the .xlsx extension in the source Folder and move it to Destination Folder

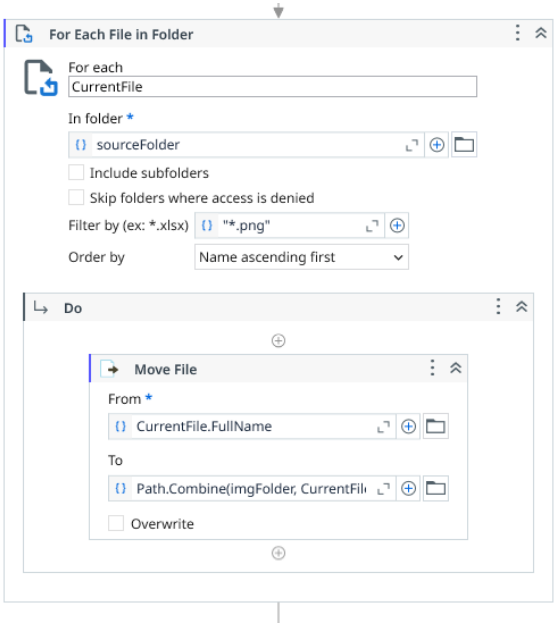


Fig 5.8 Continuation of workflow: Filter the .png extension in the source Folder and move it to Destination Folder

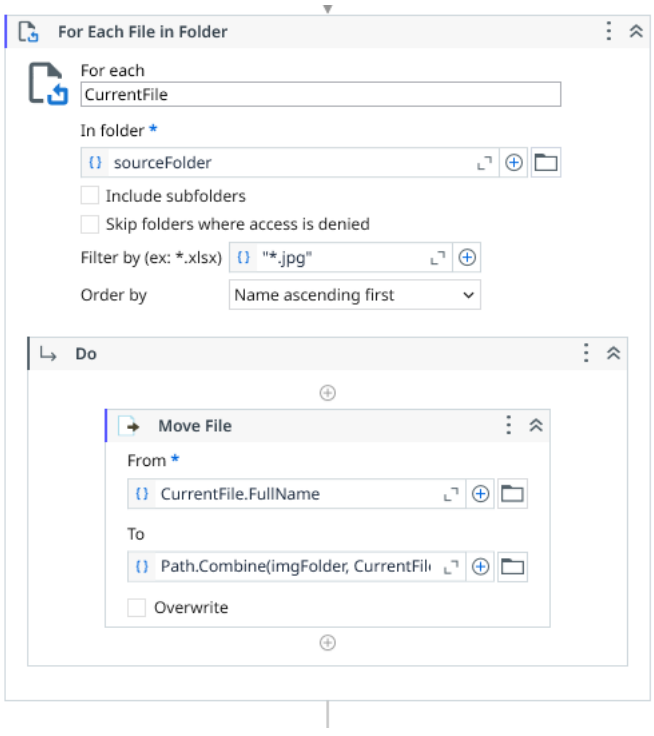


Fig 5.9 Continuation of workflow: Filter the .jpg extension in the source Folder and move it to Destination Folder

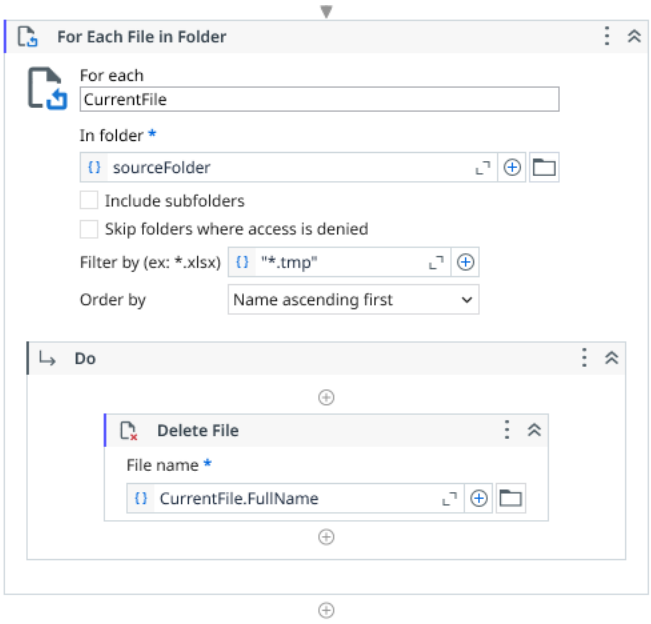


Fig 5.10 Continuation of workflow: Filter the .tmp extension in the source Folder and deletes the Files

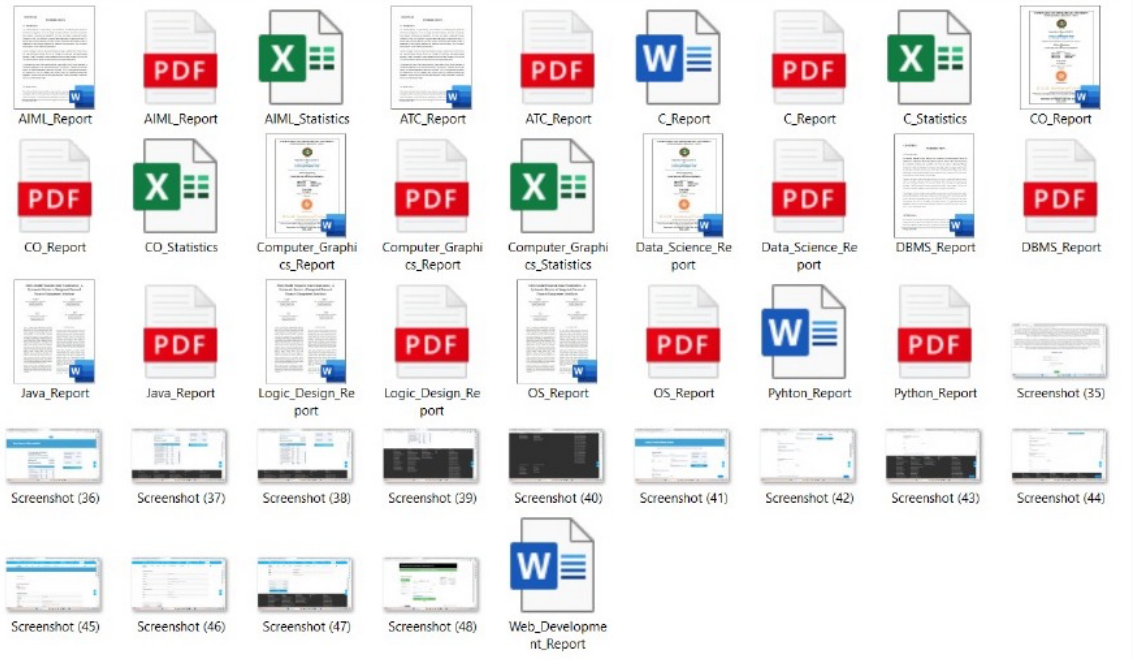


Fig 5.11 Unsorted Folder

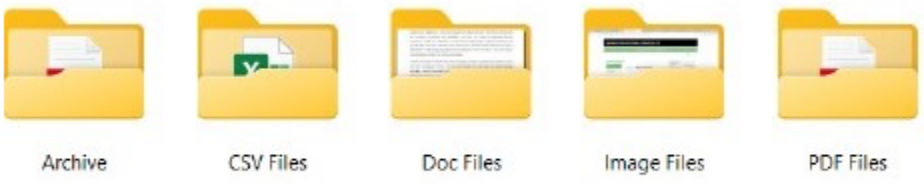


Fig 5.12 The Bot Created Sub-Folder in Source Folder

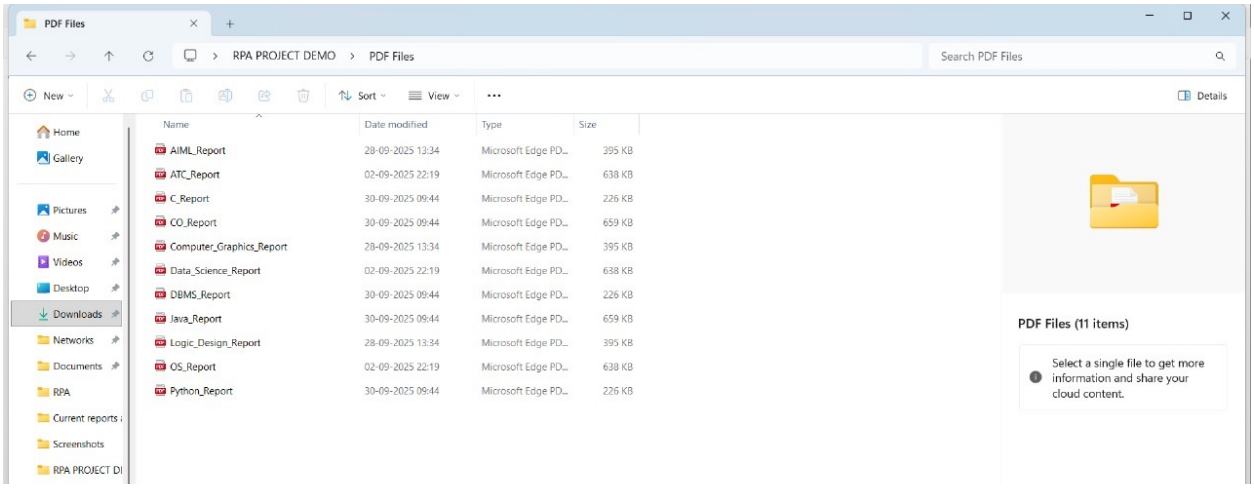


Fig 5.13 The Bot has filtered .pdf files and moved to PDF Files

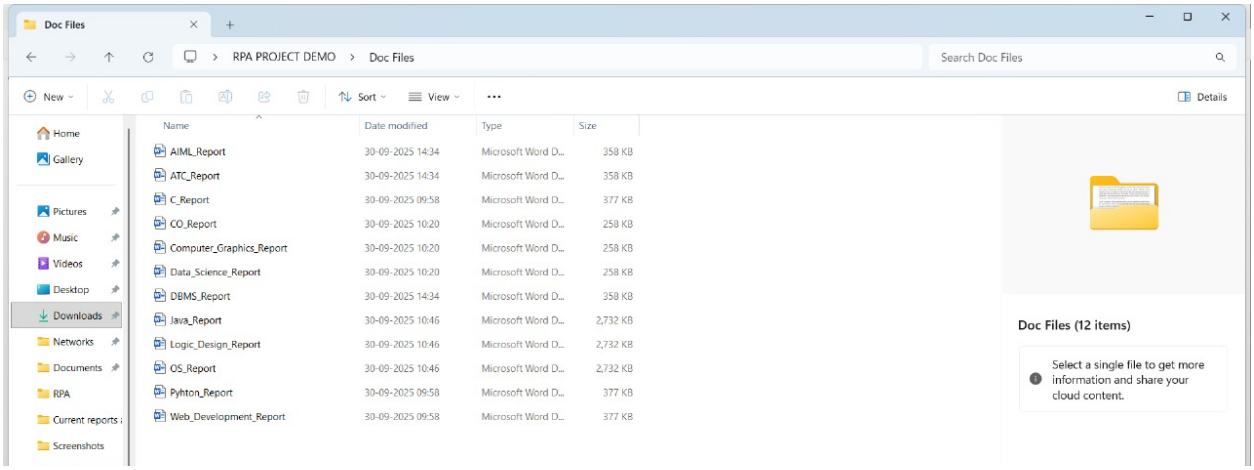


Fig 5.14 The Bot has filtered .docs files and moved to Doc Files

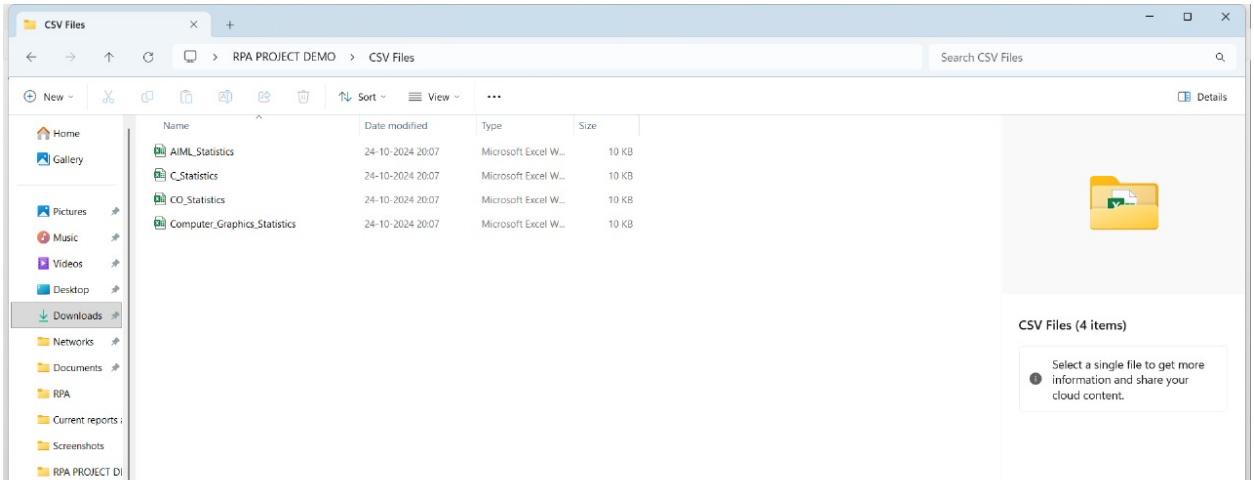


Fig 5.15 The Bot has filtered .csv and .xlsx files and moved to CSV Files

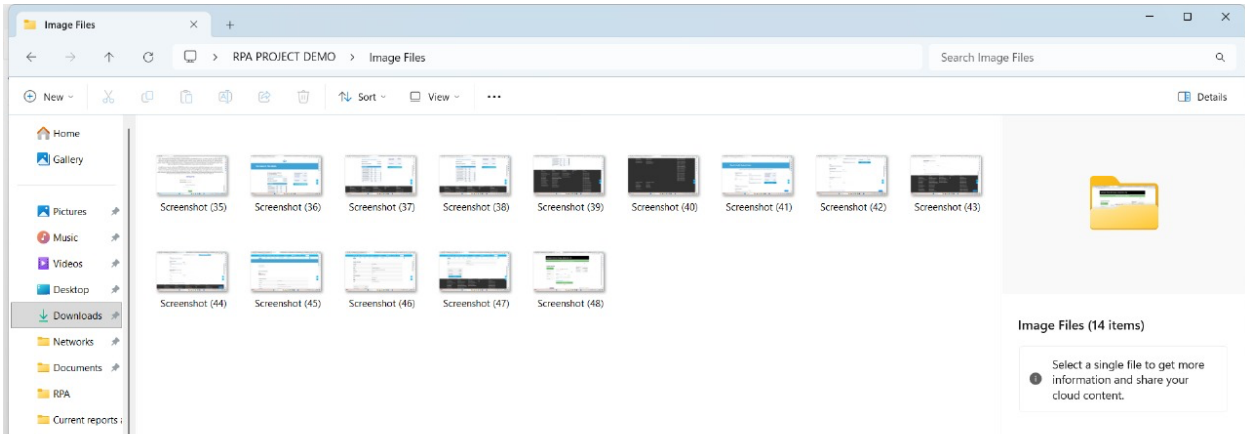


Fig 5.16 The Bot has filtered .png and .jpg files and moved to Image Files

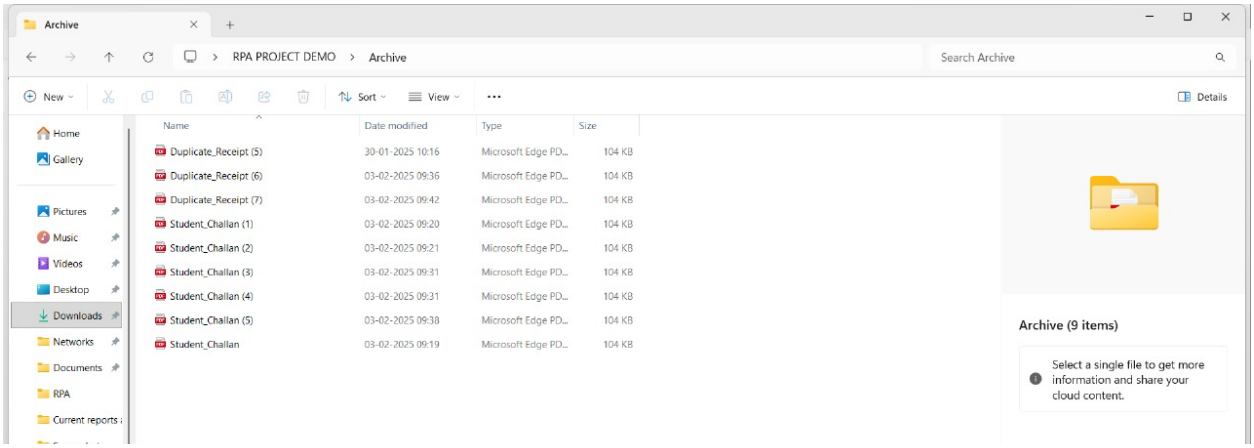


Fig 5.17 The Bot has filtered all the files un-used more the 30 days and moved to Archive Folder

CHAPTER 6

APPLICATIONS

The Automated File Organization and Cleanup System developed using UiPath has broad applicability across personal, academic, and organizational contexts. Since file management is a universal requirement, the automation can be adapted to suit different environments where digital storage and systematic organization are essential. Key applications include:

1. **Personal Use**
   - Helps individual users maintain a clean and organized Downloads or Desktop folder by automatically sorting documents, images, spreadsheets, and miscellaneous files.
   - Saves time otherwise spent manually searching and relocating files.
   - Reduces clutter and improves accessibility to important files.

2. **Academic Institutions**
   - Useful for students and researchers to organize project reports, datasets, and reference materials into designated folders.
   - Facilitates teachers in maintaining well-structured digital libraries of lecture notes, assignments, and examination-related files.
   - Prevents accumulation of outdated or duplicate academic files through automatic archiving.

3. **Business and Office Environments**
   - Assists employees in systematically managing documents such as invoices, reports, presentations, and meeting notes.
   - Prevents storage inefficiencies caused by duplicate or outdated files.
   - Enhances productivity by ensuring quicker access to the required data.

4. **Engineering and Design Teams**
   - Useful in CAD and design environments where files such as .dwg, .stl, and .pdf accumulate rapidly.

- Automates organization into separate categories like *CAD Files*, *Reports*, *Reference Docs*, etc.
- Simplifies collaboration by ensuring all team members work with a structured file system.

5. Corporate IT and Shared Folders

- Maintains digital hygiene in shared folders and team drives where multiple users store files.
- Prevents duplication and misplacement by enforcing standardized folder structures.
- Automates archiving of old project files, freeing up server space for new data.

6. Cloud and Remote Work Integration

- Can be extended to cloud-based storage solutions such as Google Drive, OneDrive, or SharePoint.
- Supports remote working scenarios where team members upload files to standard repositories that require regular cleanup.

7. General Applications

- Automated digital housekeeping for any user with frequent file downloads.
- Regularly archive unused files to save space.
- Automatic deletion of temporary and junk files to improve system performance.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The **Automated File Organization and Cleanup System** was conceptualized and implemented to solve a practical and frequently encountered problem in digital environments: accumulating and disorganizing files in commonly used folders such as Downloads or Project Files. The absence of a structured and automated mechanism for file management often results in wasted storage space, time inefficiency, and human errors in file handling. This project developed an effective solution using UiPath Studio, a leading Robotic Process Automation (RPA) platform, demonstrating how automation can improve digital hygiene and productivity in both personal and organizational contexts.

The system follows a structured methodology: it monitors a designated folder, classifies files based on their extensions, and moves them into subfolders such as *Docs*, *Reports*, and *Images*. One of the significant strengths of the solution lies in its cleanup mechanism, which identifies and archives files that have not been accessed for a specified duration (e.g., 30 days). This ensures that only relevant and active files remain in the workspace while outdated data is preserved safely in an archive folder. Additionally, the system incorporates duplicate file handling by renaming incoming files intelligently to prevent overwriting, and also removes unnecessary or temporary files (such as .tmp or .log) to maintain system cleanliness.

Practical testing of the workflow confirmed the efficiency of the automation. In scenarios where hundreds of files were accumulated, the bot successfully categorized, archived, and cleaned them within seconds, achieving significant time savings of up to 90% compared to manual effort. Furthermore, the error-free and consistent execution of the workflow highlights the reliability of RPA in automating repetitive and rule-based processes.

The project addresses file management challenges effectively and provides a learning opportunity in the domain of RPA, showcasing how UiPath's file system activities, loops, conditional logic, and exception handling can be combined into a robust automation solution. The outcome reinforces that RPA can extend beyond enterprise-level data workflows and provide valuable

applications in everyday personal computing tasks. In conclusion, the project demonstrates that automation is not merely a convenience but an essential tool for achieving efficiency, accuracy, and sustainability in modern digital ecosystems.

7.2 Future Scope

Although the project successfully meets its objectives, there is considerable scope for enhancement and scalability. The following improvements can be envisioned:

1. **Dynamic Configuration and Rule-Based Automation**
    - Currently, file categorization is defined through fixed extension rules. In the future, these rules can be stored in an external configuration file (Excel, JSON, or XML), allowing end users to easily customize file classification without editing the UiPath workflow. This would make the system more flexible and user-friendly.
2. **Cloud Integration for Hybrid Environments**
    - As cloud storage becomes the primary medium for collaboration, extending the automation to work with Google Drive, OneDrive, or SharePoint will allow seamless organization of local and cloud-stored files. This would make the system relevant for remote work and enterprise-level digital asset management.
3. **Enhanced Duplicate File Detection**
    - Presently, duplicate handling is based on filename checks. Advanced methods such as checksum comparison or content-based analysis could be integrated to detect accurate duplicates, even if the filenames differ. This would prevent unnecessary redundancy and free up storage more effectively.
4. **Notifications and Reporting**
    - Integration with UiPath.Mail.Activities or messaging APIs could allow the bot to send email/SMS notifications once a cleanup is completed. Additionally, automated Excel or PDF reports can be generated to summarize the number of files moved, archived, or deleted during each run.
5. **AI-Powered Intelligent Classification**
    - While the current system classifies files purely by extension, using Artificial Intelligence and Natural Language Processing (NLP) could allow classification

based on file content. For example, the bot could distinguish between an invoice PDF and a research article PDF and place them in separate folders.

6. **Size-Based Sorting and Priority Handling**
    - Large media files, such as videos or CAD models, can be automatically separated into dedicated folders. Additionally, priority rules can be added to ensure critical file types (e.g., project reports) are never archived without prior notification.
7. **Integration with UiPath Orchestrator for Enterprise Use**
    - The workflow can be integrated with UiPath Orchestrator for organizational deployments, enabling centralized monitoring, scheduling, and execution across multiple systems. This would transform the solution from a personal desktop tool into an enterprise-wide automation framework.
8. **Cross-Platform Expansion**
    - Although the current project is designed for Windows-based systems, future iterations could be developed for Linux and macOS environments, ensuring broader platform applicability.

The project lays the foundation for applying RPA to one of the most common challenges in digital file management. Combining automation with cleanup strategies contributes to improved productivity, better resource utilization, and a cleaner digital environment. With enhancements such as AI-based categorization, cloud integration, and enterprise deployment, the system can evolve into a comprehensive intelligent file management solution adaptable for individuals and large organizations.

REFERENCES

[1]. UiPath Documentation – File and Folder Activities, UiPath Official Guide, 2023. Available at: https://docs.uipath.com

[2]. Microsoft Docs – System.IO Namespace, Microsoft Developer Network, 2022.

[3]. Kumar, A. & Patel, R. – Application of Robotic Process Automation in Digital Workflows, International Journal of Computer Applications, Vol. 182, No. 15, 2022.

[4]. Sharma, P., Gupta, S., & Verma, R. – Automating Personal Data Organization Using Rule-Based Systems, Proceedings of the International Conference on Emerging Computing, 2020.

[5]. Automation Anywhere – Case Studies in Document and File Automation, Automation Anywhere Blog, 2021.

[6]. Blue Prism – Robotic Process Automation for Enterprise Data Management, Blue Prism Whitepaper, 2021.

[7]. File Juggler – Rule-Based File Management Tool, File Juggler Documentation, 2021.

[8]. TutorialsPoint – VB.NET File Handling, TutorialsPoint Online Resource, 2023.

[9]. Apple Inc. – Automator User Guide for macOS, Apple Developer Documentation, 2020.

[10]. VTU Course Notes – Robotic Process Automation (22CSE171), Department of Computer Science & Engineering, 2023.